Victoria Klein

Nationality British & Swiss **Languages** English (native), German (proficient) **London, UK**

victoria-klein.github.io kleinv@pm.me victoria--klein its_hapenin

Research engineer working on the evaluation of frontier models in production.

Experience

Lovable

AI Engineer, Model Evaluation — May 2026 –

- Work across the post-training and coding-agent teams evaluating new model capabilities, and identifying and correcting tail behaviours and friction between models and our harness, for Lovable's main coding agent.
- Set the team's eval methodology and standards, writing the evaluator and scoring-validity references everyone authors against. Authored core behavioural evals, such as over- and under-verification, and their judges (a null-band pairwise judge, rubric judges that must cite trajectory evidence), identifying evidence-blindness, length bias and truncated context in the house benchmark's existing judges, work recognised with Lovable's company-wide Award for Exceptional Craftsmanship.
- Designed prompt-ablation grids across effort levels and built and maintain our internal GEPA prompt-evolution engine, run against sealed holdouts, to separate model behaviour from harness-induced behaviour.
- Responsible for all internal offline evaluations of early-access checkpoints from frontier model providers, comparing them with in-house checkpoints and reporting evidence on capability, behaviour and token efficiency.
- Responsible for rollouts of newly launched models, with hourly readouts to leadership, fixing time-sensitive harness behaviour mid-rollout (Opus 5.5 build errors +237%, root-caused within 48 hours).
- Built an agentic A/B monitoring battery, a skill over deterministic scripts (SRM, Welch and z tests at collect time, exposure-aligned cohorts) that watches a rollout hourly and reports against pre-stated tripwires. Caught CUPED contamination inverting a cost result, and a production quality predictor that under-scored Claude models for editing through the shell.

Symbolica AI

AI Researcher — Sep 2024 – May 2026

- Self-initiated and implemented a SOTA approach on ARC-AGI-2 Public Eval (blog | code) using the Agentica SDK, leading to the ARC-AGI-3 approach (code) then pursued internally
- Proposed and pitched the company's first public product release with a colleague, leading the development of the Agentica SDK, an open-source framework and protocol for building and deploying "code mode" agents (blog | code | 𝕏)
- Contributed to novel encoding architectures and trained O(100M) to O(1B) parameter code synthesis models on distributed GPU infrastructure, running large-scale benchmarks including HumanEval

QUVA, University of Amsterdam

Visiting Researcher — Jan 2023 – Aug 2023

- Poster at ICML 2023 on discovering Lie group symmetries with neural networks, supervised by Efstratios Gavves

Education

Imperial College London

PhD Candidate, Mathematics — **Dec 2019 – Aug 2024**

- Supervised by Professor Jeroen S. W. Lamb, Dr Kevin N. Webster
- Part of the EPSRC CDT for Mathematics of Random Systems
- Developed a novel parametrisation of equivariant neural networks for Lie groups, implementing vectorisable and GPU-efficient computational algebra algorithms

MSc, Mathematics — **Sep 2018 – Aug 2020**

- 1st Class Honours, 83% average

Durham University

BSc, Mathematics & Computer Science — **Oct 2015 – Aug 2018**

- 1st Class Honours

Skills

Python (2015–) | Go (2026–) | TypeScript (2025–) | SQL: BigQuery, ClickHouse (2026–)
Measurement: A/B design and analysis (SRM, exposure alignment, matched pairs, cluster-robust tests), LLM-judge calibration, Braintrust, Confidence (OpenFeature), Grafana and Terraform alerting
ML: PyTorch [Flash, Flex, Lightning] (2019–) | JAX [Flax, NNX, Equinox] (2021–) | Pallas (2024–2025) | SLURM (2022–2023)
Web: React/Node.js (2025–)

Awards & Publications

2026

- Award for Exceptional Craftsmanship, Lovable (company-wide values award), for model evaluation work

2023

- ICML 2023, Workshop on Topology, Algebra and Geometry in Machine Learning (TAG-ML) | Poster (paper)
- The Fields Institute 3rd Symposium for Machine Learning and Dynamical Systems | Poster (paper)

2022

- Dorris Chen Award, Department of Mathematics, Imperial College London
- The Fields Institute Travel Grant, The Fields Institute

Talks, Projects & Community

Talks

- Navigate 2026 (Browserbase, San Francisco): "Browsers for Coding Agents: The Two Tails of Verification"
- ARC-AGI-2 at Dust.tt Engineering Night and AI Nerd meetup

Personal — victoria-klein.github.io/projects

- JAX/NumPy implementation of the Growing Basis Algorithm
- Synthesis of JAX NNX Graphs from PyTreeDefs

Teaching

- Graduate tutorials in DL (2021–2023); undergraduate differential equations (2022) and calculus (2021), Imperial

Community

- Code First Girls Fellow (2020)
- Mary Lister McCammon Fellowship Talk, Imperial (2020 & 2021)
- Committee, WomenInStem@IC Society (2020) | Women in Maths Society (2021), Imperial